

SQL Q&A

1. What is SQL?

Answer:

- SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases. It is used to query, insert, update, and delete data from a database.
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2. What are the different types of SQL commands?

Answer:

- **DML (Data Manipulation Language):** SELECT, INSERT, UPDATE, DELETE.
 - **DDL (Data Definition Language):** CREATE, ALTER, DROP, TRUNCATE.
 - **DCL (Data Control Language):** GRANT, REVOKE.
 - **TCL (Transaction Control Language):** COMMIT, ROLLBACK, SAVEPOINT.
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3. What is the difference between DELETE, TRUNCATE, and DROP?

Answer:

- **DELETE:** Removes rows from a table based on a condition; can be rolled back.
 - **TRUNCATE:** Removes all rows from a table; faster but cannot be rolled back.
 - **DROP:** Removes the entire table or database from the system.
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4. What is a primary key?

Answer:

- A **primary key** is a field or combination of fields in a table that uniquely identifies each record. It cannot contain NULL values and must be unique.
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5. What is a foreign key?

Answer:

- A **foreign key** is a field in one table that uniquely identifies a row of another table. It establishes a link between two tables by enforcing referential integrity.
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6. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?

Answer:

- **INNER JOIN:** Returns records that have matching values in both tables.
 - **LEFT JOIN (or LEFT OUTER JOIN):** Returns all records from the left table and matching records from the right table.
 - **RIGHT JOIN (or RIGHT OUTER JOIN):** Returns all records from the right table and matching records from the left table.
 - **FULL OUTER JOIN:** Returns all records when there is a match in either left or right table.
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7. What is the GROUP BY clause in SQL?

Answer:

- The **GROUP BY** clause is used to group rows that have the same values into summary rows, like finding the sum, average, or count for each group.
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8. What is a subquery?

Answer:

- A **subquery** is a query nested inside another query. It is used to retrieve intermediate results that can be used by the main query.
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9. What is a JOIN in SQL?

Answer:

- A **JOIN** is used to combine rows from two or more tables based on a related column. Common types of joins include INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN.

10. What is the difference between UNION and UNION ALL?

Answer:

- **UNION:** Combines the result sets of two queries and removes duplicates.
 - **UNION ALL:** Combines the result sets of two queries, including duplicates.
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11. What is an index in SQL?

Answer:

- An **index** is a database object that improves the speed of data retrieval operations. It creates a faster lookup for queries that use columns in the WHERE clause or join conditions.
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12. What is normalization in SQL?

Answer:

- **Normalization** is the process of organizing data in a database to reduce redundancy and dependency. The goal is to separate data into different tables to minimize duplication.
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13. What are the different normal forms in SQL?

Answer:

- **1st Normal Form (1NF):** Ensures all columns contain atomic values (no repeating groups).
 - **2nd Normal Form (2NF):** Achieves 1NF and removes partial dependency.
 - **3rd Normal Form (3NF):** Achieves 2NF and removes transitive dependency.
 - **Boyce-Codd Normal Form (BCNF):** A stricter version of 3NF.
 - **4th and 5th Normal Form (4NF and 5NF):** Deals with multi-valued dependencies and join dependencies.
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14. What is denormalization in SQL?

Answer:

- **Denormalization** is the process of combining tables to reduce the number of joins needed in queries, improving read performance at the expense of increased data redundancy and write complexity.
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15. What is the HAVING clause in SQL?

Answer:

- The **HAVING** clause is used to filter the results of a GROUP BY query based on an aggregate condition (e.g., COUNT, SUM, AVG).
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16. What is a CASE statement in SQL?

Answer:

- The **CASE** statement is used for conditional logic in SQL. It allows you to return a value based on conditions.

```
SELECT id,  
       CASE  
         WHEN age >= 18 THEN 'Adult'  
         ELSE 'Minor'  
       END AS age_group  
FROM persons;
```

17. What is the DISTINCT keyword in SQL?

Answer:

- The **DISTINCT** keyword is used to remove duplicate rows from the result set.
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18. What is the difference between WHERE and HAVING?

Answer:

- **WHERE** filters rows before grouping.
 - **HAVING** filters rows after grouping (usually used with aggregate functions).
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19. What is the purpose of LIMIT or TOP in SQL?

Answer:

- **LIMIT** (in MySQL and PostgreSQL) or **TOP** (in SQL Server) is used to restrict the number of rows returned by a query.

```
SELECT * FROM employees LIMIT 5;
```

20. What is an AUTO_INCREMENT in SQL?

Answer:

- **AUTO_INCREMENT** is a property used in MySQL to automatically increment the value of a field, often used for primary keys.
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21. What are the aggregate functions in SQL?

Answer:

- **COUNT**: Counts the number of rows.
 - **SUM**: Sums the values of a column.
 - **AVG**: Averages the values of a column.
 - **MIN**: Finds the minimum value in a column.
 - **MAX**: Finds the maximum value in a column.
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22. What is a view in SQL?

Answer:

- A **view** is a virtual table based on the result of a query. It can simplify complex queries and present data in a specific way.

23. What is the ALTER command in SQL?

Answer:

- The **ALTER** command is used to modify the structure of an existing database object, such as a table, by adding or deleting columns, or changing data types.

24. What is the TRUNCATE command in SQL?

Answer:

- **TRUNCATE** removes all rows from a table, but the table structure and schema remain intact. It is faster than DELETE because it doesn't log individual row deletions.

25. What are stored procedures in SQL?

Answer:

- **Stored procedures** are precompiled collections of SQL statements that can be executed as a single unit. They help improve performance and reuse SQL code.

26. What is the EXPLAIN keyword in SQL?

Answer:

- **EXPLAIN** provides information about how the SQL engine executes a query, including the execution plan and indexes used.

27. What is the difference between CHAR and VARCHAR?

Answer:

- **CHAR**: A fixed-length string.
- **VARCHAR**: A variable-length string, which only uses as much space as needed.

28. What is a self-join in SQL?

Answer:

- A **self-join** is a join where a table is joined with itself. This is useful when you need to compare rows within the same table.
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29. What is the purpose of INDEX in SQL?

Answer:

- An **index** improves the speed of data retrieval operations on a table. It is created on columns that are frequently used in WHERE, JOIN, or ORDER BY clauses.
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30. What is the IN operator in SQL?

Answer:

- The **IN** operator is used to filter the result set based on a list of values.

```
SELECT * FROM employees WHERE department IN ('HR', 'Finance');
```

31. What is the BETWEEN operator in SQL?

Answer:

- The **BETWEEN** operator is used to filter data within a specific range, including the boundary values.

```
SELECT * FROM products WHERE price BETWEEN 50 AND 100;
```

32. What is the difference between RANK() and DENSE_RANK()?

Answer:

- **RANK()**: Assigns the same rank to equal values, but leaves gaps in the ranking.
 - **DENSE_RANK()**: Assigns the same rank to equal values, without leaving gaps.
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33. What is the COALESCE() function in SQL?

Answer:

- The **COALESCE()** function returns the first non-NULL value in a list of arguments.

```
SELECT COALESCE(NULL, NULL, 'Hello', 'World');
```

34. What is the difference between AND and OR in SQL?

Answer:

- **AND:** Returns true if both conditions are true.
 - **OR:** Returns true if either of the conditions is true.
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35. How do you find duplicate records in a table?

Answer:

- Use the GROUP BY clause along with HAVING to find duplicates.

```
SELECT column_name, COUNT(*) FROM table_name GROUP BY column_name HAVING  
COUNT(*) > 1;
```

36. What is a composite key?

Answer:

- A **composite key** is a primary key that consists of more than one column in order to uniquely identify a record.
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37. What is the INTERSECT operator in SQL?

Answer:

- The **INTERSECT** operator returns the common records between two SELECT queries.
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38. What is NULL in SQL?

Answer:

- **NULL** represents the absence of a value or unknown data. It is not the same as an empty string or zero.

39. What is the difference between INNER JOIN and OUTER JOIN?

Answer:

- **INNER JOIN:** Returns records with matching values from both tables.
- **OUTER JOIN:** Returns all records from one table, and matching records from the other.

40. How do you optimize SQL queries?

Answer:

- Use indexing, avoid SELECT *, minimize the use of JOIN operations, and ensure that WHERE conditions are efficient.